Tutorial #4

Section 2.1

2.) Using set builder notation → there are infinite number of correct answers.

a). $\{3n \mid n = 0, 1, 2, 3, 4\}$

b). Assume the domain is the set of integers

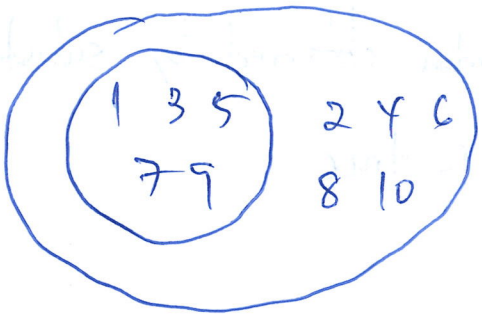
$\{x \mid -3 < x \leq 3\}$

c). $\{x \mid x$ is a letter of the word monopoly other than l or y$\}$

4.)

$B \subseteq A$	$A \subseteq A$
$C \subseteq A$	$B \subseteq B$
$C \subseteq D$	$C \subseteq C$

12.) The numbers 1, 3, 5, 7, and 9 form a subset of the set of all ten positive integers under discussion.



32) $A = \{a, b, c\}$, $B = \{x, y\}$ and $C = \{0, 1\}$

a). $A \times B \times C = \{(a, x, 0), (a, x, 1), (a, y, 0), (a, y, 1),$
$(b, x, 0), (b, x, 1), (b, y, 0), (b, y, 1),$
$(c, x, 0), (c, x, 1), (c, y, 0), (c, y, 1)\}$

b). $C \times B \times A = \{(0,x,a),(0,x,b),(0,x,c),(0,y,a),(0,y,b),(0,y,c),$
$(1,x,a),(1,x,b),(1,x,c),(1,y,a),(1,y,b),(1,y,c)\}$

c). $C \times A \times B = \{(0,a,x),(0,a,y),(0,b,x),(0,b,y),(0,c,x),(0,c,y),$
$(1,a,x),(1,a,y),(1,b,x),(1,b,y),(1,c,x),(1,c,y)\}$

d). $B \times B \times B = \{(x,x,x),(x,x,y),(x,y,x),(x,y,y),$
$(y,x,x),(y,x,y),(y,y,x),(y,y,y)\}$

34). a). $\exists x \in \mathbb{R}(x^3=-1)$

There is a real whose cube is -1. True, $x=-1$ is a solution.

b) $\exists x \in \mathbb{Z}(x+1>x)$

There is an integer such that the number obtained by adding 1 to it is greater than the integer.

This is true.

c). $\forall x \in \mathbb{Z}(x-1 \in \mathbb{Z})$

For every integer, the number obtained by substracting 1 is an integer. This is true.

d). $\exists x \in \mathbb{Z}(x^2 \in \mathbb{Z})$

The square of every integer is an integer. This is true.

Section 2.2 Set Operation.

4). Note that $A \subseteq B$

a). $A \cup B = \{a, b, c, d, e, f, g, h\} = B$

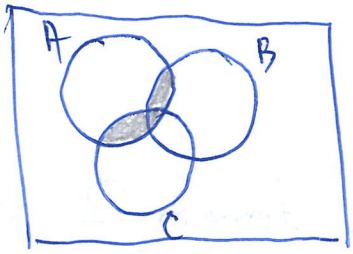
b). $A \cap B = \{a, b, c, d, e\} = A$

c). $A - B = \emptyset$

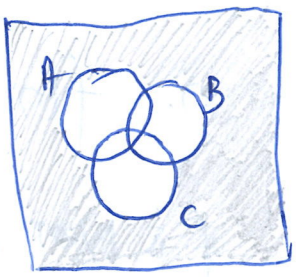
d). $B - A = \{f, g, h\}$

26). Venn diagram

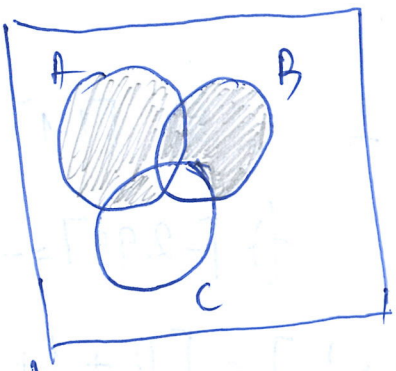
a). $A \cap (B \cup C)$



b). $\bar{A} \cap \bar{B} \cap \bar{C}$



c). $(A - B) \cup (A - C) \cup (B - C)$



52). $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

a). $\{3, 4, 5\} \rightarrow$ 00 1110 0000

b). $\{1, 3, 6, 10\} \rightarrow$ 10 1001 0001

c). $\{2, 3, 4, 7, 8, 9\} \rightarrow$ 01 1100 1110

Section 2.3 Functions

2) a). From Definition #1, this is not a function, $(f(n) = \pm n)$

e.g. $f(2) = 2$ or $f(2) = -2$

2 is mapped to 2 and -2 (both at the same time).

$$2 \to 2$$
$$2 \to -2$$

b). $f(n) = \sqrt{n^2+1}$

f is a function from $\mathbb{Z}$ to $\mathbb{R}$, for all integers n, $\sqrt{n^2+1}$ is a well-defined real number.

c). $f(n) = \dfrac{1}{n^2-4}$

f is NOT a function with domain $\mathbb{Z}$ because $n = \pm 2$, $f(n)$ is not defined.

8) a). $\lfloor 1.1 \rfloor = 1$ b) $\lceil 1.1 \rceil = 2$ c). $\lfloor -0.1 \rfloor = -1$ d). $\lceil 0.1 \rceil = 0$

e). $\lceil 2.999 \rceil = 3$ f). $\lceil -2.999 \rceil = -2$ g). $\lfloor \frac{1}{2} + \lceil \frac{1}{2} \rceil \rfloor = \lfloor \frac{1}{2} + 1 \rfloor = \lfloor \frac{3}{2} \rfloor = 1$

h). $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil = \lceil 0 + 1 + \frac{1}{2} \rceil = \lceil \frac{3}{2} \rceil = 2$

12). $\mathbb{Z}$ to $\mathbb{Z}$

a). $f(n) = n-1$. This is one-to-one

Definition #5, $f(a) = f(b)$ implies that $a = b$

$f(a) = a-1$

$f(b) = b-1$, if $f(a) = f(b) \Rightarrow a-1 = b-1$ or $a = b$ true.

b). $f(n) = n^2+1$, not one-to-one e.g. $f(2) = f(-2) = 5$

c). $f(n) = n^3$

Definition #5, $f(a) = a^3$
$f(b) = b^3$

$f(a) = f(b) \Rightarrow a = b$ true, this is one-to-one.

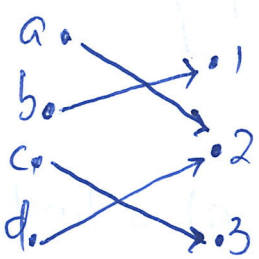
d). $f(n) = \lceil \frac{n}{2} \rceil$

If $n = 3$, $f(3) = \lceil \frac{3}{2} \rceil = 2$

$n = 4$, $f(4) = \lceil \frac{4}{2} \rceil = 2$, this is not one-to-one

14). $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is it onto?

e.g.



a). $f(m,n) = 2m - n$ (2 variables, one equation)

For any $k \in \mathbb{Z}$

if $m = k$, $n = k$

$f(m,n) = f(k,k) = 2k - k = k$. This is on-to.

or if $m = 0$, $f(m,n) = -n$

We have $n \in \mathbb{Z}$, $f(m,n) = -n \in \mathbb{Z}$. This is onto.
(Every number is in the range)

b). $f(m,n) = m^2 - n^2$

$= (m-n)(m+n)$

~~If m is odd & n is even~~ $\Rightarrow$ ~~$m = 2k_1 + 1$~~, ~~$n = 2k_2$~~ $\Rightarrow$ ~~$m - n = 2(k_1 - k_2) + 1$~~

~~$(m-n)(m+n) = 4(k_1^2 - k_2^2) + 4k_1 + 1$~~ ~~$m + n = 2(k_1 + k_2) + 1$~~

2 is not in the range because $2 = 1 \cdot 2$ or $(-1)(-2)$

$(m-n)$ and $(m+n)$ must be 1 and 2 or 2 and 1
or -1 and -2 or -2 and -1

$$\begin{cases} m-n=1 \\ m+n=2 \end{cases} \quad \text{or} \quad \begin{cases} m-n=2 \\ m+n=1 \end{cases} \quad \text{or} \quad \begin{cases} m-n=-1 \\ m+n=-2 \end{cases} \quad \begin{matrix} m-n=-2 \\ m+n=-1 \end{matrix}$$

$$m=\frac{3}{2} \notin \mathbb{Z} \qquad m=\frac{3}{2} \notin \mathbb{Z} \qquad m=-\frac{3}{2} \notin \mathbb{Z} \qquad m=-\frac{3}{2} \in \mathbb{Z}$$

We cannot produce 2 as the output for $m \& n \in \mathbb{Z} \times \mathbb{Z}$

$\therefore$ The function is not onto. (2 is not in the range)

c). $f(m,n) = m+n+1$

$\left.\begin{matrix} m=0 \\ n=k-1 \end{matrix}\right\} \Rightarrow f(0,k-1)=k$ for every integer k.

Cover all $\therefore$ This function is ONTO.

d). $f(m,n) = |m|-|n|$.

To cover negative values we set $m=0$,

To cover/achieve nonnegative values we set $n=0$.

$\therefore$ ONTO

e). $f(m,n)=m^2-4$. This function does NOT depend on n at all.

To mak it onto, we must have

$\forall k \in \mathbb{Z}, \exists m,n \in \mathbb{Z}$ such that $m^2-4=k$

or $m^2=k+1$

If $k=-5$ then $m^2=-5+1=-1$

$\therefore k=-5$ or -5 is NOT in the range

$\therefore$ NOT ONTO.

36). $f(x) = x^2+1$

$g(x) = x+2$

$(f \circ g)(x) = f[g(x)] = f(x+2) = (x+2)^2+1 = x^2+4x+5$

$(g \circ f)(x) = g[f(x)] = g(x^2+1) = x^2+1+2 = x^2+3$.